

EXACT POST-SELECTION INFERENCE, WITH APPLICATION TO THE LASSO

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We develop an approach to performing valid inference after model selection. In a nutshell, our approach gives post-selection inferences with the same frequency interpretation as those given by data/sample splitting, but it is more broadly applicable and more powerful. The approach does *not* depend on the correctness of the selected model; *i.e.* it gives valid inferences even when the selected model is incorrect. At the core of our framework is a result that characterizes the distribution of a post-selection estimator conditioned on the *selection event*. We specialized the approach to model selection by the lasso to form valid confidence intervals for the selected coefficients and test whether all relevant variables have been included in the model.

1. Introduction. As a statistical technique, linear regression is both simple and powerful. Not only does it provide estimates of the “effect” of each variable, but it also quantifies the uncertainty in those estimates, allowing for confidence intervals and hypothesis tests of the effect sizes. However, in many applications, a practitioner starts with a large pool of candidate variables, such as genes or demographic features, and does not know *a priori* which are relevant. This problem is especially acute if there are more variables than observations, when the linear model is unidentified.

A practitioner might wish to use the data to select a model and then perform inference on the selected model. As an example, suppose she fits the linear model, observes which coefficients are significant at level α , then reports the p -values for those coefficients. Intuitively, this is problematic because she selects predictors that happened to be significant on this instance of the data, so if she uses the same data for inference, the p -values she reports will all tend to be small, including for null effects.

To remedy this problem, we first have to formalize the goals of a post-selection procedure. This is easier said than done because in linear regression, the target of inference changes with the selected model. To see this, suppose

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the responses $\mathbf{y} \in \mathbb{R}^n$ are generated according to

$$(1.1) \quad \mathbf{y} \sim N(\boldsymbol{\mu}, \sigma^2 I_n),$$

and we wish to model $\boldsymbol{\mu}$ as a linear function of predictors $\mathbf{x}_1, \dots, \mathbf{x}_p \in \mathbb{R}^n$. We choose a subset $M \subset \{1, \dots, p\}$ and ask for the linear combination of the predictors in M that minimize the expected error, i.e.,

$$(1.2) \quad \boldsymbol{\beta}^M = \arg \min_{\mathbf{b}} \mathbb{E} \|\mathbf{y} - X_M \mathbf{b}\|^2 = X_M^+ \boldsymbol{\mu},$$

where $X_M^+ \equiv (X_M^T X_M)^{-1} X_M^T$ is the pseudo-inverse of X_M . Notice that even for the same predictor j , the targets β_j^M and $\beta_j^{M'}$ in different models M and M' will in general be different. This reflects the well-known fact that regression coefficients describe the effect of a predictor, *adjusting for the other predictors in the model*.

Thus, in linear regression, the target changes with the model M (Berk et al., 2013). In order to make sense of post-selection inference, we first need to define what it means to do inference for a “moving” target. As we see in the next section, there are several reasonable choices.

2. Post-Selection Inference in Linear Regression. Although the target in linear regression appears to change with the model, it is not so much the target that is changing as our choice of which targets to consider. In other words, we can imagine that there are $p2^{p-1}$ well-defined population parameters, one for each coefficient in all 2^p possible models, i.e.,

$$\{\beta_j^M : M \subset \{1, \dots, p\}, j \in M\},$$

and when we select a model $\hat{M}$, we perform inference on a subset of these.

What guarantees should a post-selection inference procedure have? The obvious choice, to control $\mathbb{P}(\beta_j^{\hat{M}} \in C_j^{\hat{M}})$, does not make sense because $\beta_j^{\hat{M}}$ does not exist when $j \notin \hat{M}$. Berk et al. (2013) note that there are essentially two options:

1. Conditional Coverage: Since we form an interval for β_j^M if and only if we select model M , i.e., $\hat{M} = M$, it makes sense to condition on this event. Hence, we might require that our confidence interval C_j^M satisfy

$$(2.1) \quad \mathbb{P}(\beta_j^M \in C_j^M \mid \hat{M} = M) \geq 1 - \alpha.$$

The benefit of this approach is that we avoid having to ever compare coefficients across different models M and M' .

2. Simultaneous Coverage: We can make sense of events and random variables that are defined simultaneously over all $j \in \hat{M}$. [Berk et al. \(2013\)](#) propose controlling familywise error rate (FWER)

$$(2.2) \quad \mathbb{P}(\beta_j^{\hat{M}} \notin C_j^{\hat{M}} \text{ for any } j \in \hat{M}) \leq \alpha,$$

but one can just as well control the false coverage rate (FCR) of [Benjamini and Yekutieli \(2005\)](#):

$$(2.3) \quad FCR = \mathbb{E} \left[\frac{\#\{j \in \hat{M} : \beta_j^{\hat{M}} \notin C_j^{\hat{M}}\}}{\#\hat{M}} \right] \leq \alpha.$$

Remarkably, if we form conditional intervals (2.1) for each coefficient in our model $\hat{M}$, then this family of intervals controls FCR (2.3).

THEOREM 2.1. *Suppose we have a procedure that produces a family of intervals $\{C_j^{\hat{M}}\}_{j \in \hat{M}}$ that each have conditional $(1 - \alpha)$ coverage:*

$$\mathbb{P}(\beta_j^{\hat{M}} \notin C_j^{\hat{M}} | \hat{M} = M) \leq \alpha, \text{ for all } M \text{ and } j \in M.$$

Then, $FCR \leq \alpha$.

PROOF. We condition on $\hat{M}$ and iterate expectations.

$$\begin{aligned} FCR &= \mathbb{E} \left[\mathbb{E} \left[\frac{\#\{j \in \hat{M} : \beta_j^{\hat{M}} \notin C_j^{\hat{M}}\}}{\#\hat{M}} \middle| \hat{M} \right] \right] \\ &= \mathbb{E} \left[\frac{\mathbb{E} [\#\{j \in \hat{M} : \beta_j^{\hat{M}} \notin C_j^{\hat{M}}\} | \hat{M}]}{\#\hat{M}} \right] \\ &= \mathbb{E} \left[\frac{\sum_{j \in \hat{M}} \mathbb{P}(\beta_j^{\hat{M}} \notin C_j^{\hat{M}} | \hat{M})}{\#\hat{M}} \right] \\ &\leq \mathbb{E} \left[\frac{\alpha |\hat{M}|}{|\hat{M}|} \right] \\ &= \alpha. \end{aligned}$$

□

3. Outline of Our Approach. We have argued that

$$\mathbb{P}(\beta_j^M \in C_j^M \mid \hat{M} = M) \geq 1 - \alpha.$$

should be the goal of post-selection inference in regression, both because conditional inference is interesting in its own right and because it implies FCR control. We develop a way to control this by studying, more generally, the conditional distribution

$$(3.1) \quad \boldsymbol{\eta}_M^T \mathbf{y} \mid \{\hat{M} = M\}$$

to form inferences for $\boldsymbol{\eta}_M^T \boldsymbol{\mu}$. Notice that $\beta_j^M = e_j^T X_M^+ \boldsymbol{\mu}$ can be written in this form $\boldsymbol{\eta}_M^T \boldsymbol{\mu}$, as can many other linear contrasts; our framework is not restricted to inference for β_j^M .

Since our paper focuses on the specific case where the lasso is used for model selection, we begin in Section 4 by characterizing the set $\{\hat{M} = M\}$ for the lasso. As it turns out, this set is a union of polytopes. More precisely, the set $\{\hat{M} = M, \hat{\mathbf{s}}_M = \mathbf{s}_M\}$, that specifies the model *and* the signs of the selected variables, is a polytope of the form

$$\{A(M, \mathbf{s}_M) \mathbf{y} \leq \mathbf{b}(M, \mathbf{s}_M)\}.$$

Therefore, if we condition on both the model and the signs, we can study the simpler problem of

$$(3.2) \quad \boldsymbol{\eta}^T \mathbf{y} \mid \{A \mathbf{y} \leq \mathbf{b}\}.$$

We do this in Section 5. It turns out that this conditional distribution is “morally” a (univariate) truncated Gaussian. We use this to derive a pivot F whose distribution given $\{A \mathbf{y} \leq \mathbf{b}\}$ is $\text{Unif}(0, 1)$.

3.1. *Related Work.* CITE CHERNO’S PAPERS

The resulting post-selection test has a similar structure to the pathwise significance tests of Lockhart et al. (2014) and Taylor et al. (2014), which also are conditional tests. However, their intended application is different. They test the specific hypothesis of whether each newly added coefficient along the LARS path is relevant. By contrast, our framework allows for tests at any fixed value of λ . Furthermore, we can ask more general questions about the lasso-selected model—specifically, inference about individual coefficients in the model, which is often what practitioners want to know.

In terms of intended application, our method is more similar to work on inference in high-dimensional regression based on “de-biasing” the lasso estimator (Javanmard and Montanari, 2013; van de Geer et al., 2013; Zhang

and Zhang, 2014). These works also address forming confidence intervals for individual coefficients, except that the target is β^0 , the coefficients in the full model. Our approach differs from theirs in that our targets are the coefficients $\beta_{E,j}$ in the selected model E . This will differ from β_j^0 unless E contains all non-zero coefficients of β^0 . Although inference for β^0 is appealing, it requires assumptions about the correctness of the linear model and the sparsity of β^0 . We share the view of Berk et al. (2013) that it is generally more appropriate to think of the model we select as a linear approximation to the truth and perform inference based on this.

4. The Lasso and Its Selection Event. In this paper, we apply our post-selection inference procedure to the model selected by the lasso (Tibshirani, 1996). The lasso estimate is the solution to the usual least squares problem with an additional ℓ_1 penalty on the coefficients:

$$(4.1) \quad \hat{\beta} \in \arg \min_{\beta} \frac{1}{2} \|y - X\beta\|_2^2 + \lambda \|\beta\|_1.$$

The ℓ_1 penalty shrinks many of the coefficients to exactly zero, and the tradeoff between sparsity and fit to the data is controlled by the penalty parameter $\lambda \geq 0$.

Because the lasso produces sparse solutions, we can define model “selected” by the lasso to be simply the predictors with non-zero coefficients:

$$\hat{M} = \{j : \hat{\beta}_j \neq 0\}.$$

Then, post-selection inference seeks to make inferences about β^M , conditional on $\hat{M} = M$, as defined in (1.2).

The rest of this section will focus on characterizing the event $\{\hat{M} = M\}$. We begin by noting that in order for a vector of coefficients $\hat{\beta}$ and a vector of signs $\hat{s}$ to be solutions to the lasso problem (4.1), it is necessary and sufficient that they satisfy the Karush-Kuhn-Tucker (KKT) conditions:

$$(4.2) \quad X^T(X\hat{\beta} - y) + \lambda\hat{s} = 0,$$

$$\hat{s}_i = \text{sign}(\hat{\beta}_j) \quad \text{if } \hat{\beta}_j \neq 0$$

$$(4.3) \quad \hat{s}_i \in [-1, 1] \quad \text{if } \hat{\beta}_j = 0$$

Following Tibshirani (2013), we consider the *equicorrelation set*

$$(4.4) \quad \hat{E} = \{i \in \{1, \dots, p\} : |\hat{s}_i| = 1\}.$$

Since $\hat{s}_i \in \{-1, 1\}$ for any $\hat{\beta}_i \neq 0$, $\hat{E} \supset \hat{M}$. Although $\hat{E}$ may in principle contain predictors whose coefficients are zero, this happens with probability

zero for almost every λ . Hence, $\hat{E} = \hat{M}$ for all practical purposes, so we focus on characterizing $\{\hat{E} = E\}$.

It turns out that it is easier to first characterize $\{(\hat{E}, \hat{\mathbf{s}}) = (E, \mathbf{s})\}$, and then obtain $\{\hat{E} = E\}$ as corollary by taking a union over the possible signs. The next result is an important first step.

LEMMA 4.1. *Assume the columns of X are in general position (Tibshirani, 2013). Let $E \subset \{1, \dots, p\}$ and $\mathbf{s} \in \{-1, 1\}^{|E|}$ be a candidate set of variables and their signs, respectively. Define the random variables*

$$(4.5) \quad \mathbf{w}(E, \mathbf{s}) := (X_E^T X_E)^{-1} (X_E^T \mathbf{y} - \lambda \mathbf{s})$$

$$(4.6) \quad \mathbf{u}(E, \mathbf{s}) := X_{-E}^T (X_E^T)^+ \mathbf{s} + \frac{1}{\lambda} X_{-E}^T (I - P_E) \mathbf{y}.$$

Then the selection procedure can be rewritten in terms of $\mathbf{w}$ and $\mathbf{u}$ as:

$$(4.7) \quad \{(\hat{E}, \hat{\mathbf{s}}) = (E, \mathbf{s})\} = \left\{ \text{sign}(\mathbf{w}(E, \mathbf{s})) = \mathbf{s}, \|\mathbf{u}(E, \mathbf{s})\|_\infty < 1 \right\}$$

PROOF. First, we rewrite the KKT conditions (4.2) by partitioning them according to the equicorrelation set $\hat{E}$, adopting the convention that $-\hat{E}$ means “variables not in $\hat{E}$.”

$$\begin{aligned} X_{\hat{E}}^T (X_{\hat{E}} \hat{\boldsymbol{\beta}}_{\hat{E}} - \mathbf{y}) + \lambda \hat{\mathbf{s}}_{\hat{E}} &= 0 \\ X_{-\hat{E}}^T (X_{\hat{E}} \hat{\boldsymbol{\beta}}_{\hat{E}} - \mathbf{y}) + \lambda \hat{\mathbf{s}}_{-\hat{E}} &= 0 \\ \text{sign}(\hat{\boldsymbol{\beta}}_{\hat{E}}) &= \hat{\mathbf{s}} \\ \|\hat{\mathbf{s}}_{-\hat{E}}\|_\infty &< 1. \end{aligned}$$

Since the KKT conditions are necessary and sufficient for a solution, we obtain that $\{(\hat{E}, \hat{\mathbf{s}}) = (E, \mathbf{s})\}$ if and only if there exist $\mathbf{w}$ and $\mathbf{u}$ satisfying:

$$\begin{aligned} X_E^T (X_E \mathbf{w} - \mathbf{y}) + \lambda \mathbf{s} &= 0 \\ X_{-E}^T (X_E \mathbf{w} - \mathbf{y}) + \lambda \mathbf{u} &= 0 \\ \text{sign}(\mathbf{w}) &= \mathbf{s} \\ \|\mathbf{u}\|_\infty &< 1. \end{aligned}$$

We can solve the first two equations for $\mathbf{w}$ and $\mathbf{u}$ to see that they must satisfy

$$\begin{aligned} \mathbf{w} &= (X_E^T X_E)^{-1} (X_E^T \mathbf{y} - \lambda \mathbf{s}) \\ \mathbf{u} &= X_{-E}^T (X_E^T)^+ \mathbf{s} + \frac{1}{\lambda} X_{-E}^T (I - P_E) \mathbf{y} \\ \text{sign}(\mathbf{w}) &= \mathbf{s} \\ \|\mathbf{u}\|_\infty &< 1. \end{aligned}$$

The first two “conditions” are definitions (4.5) and (4.6), and the last two are the necessary and sufficient conditions (4.7). $\square$

Lemma 4.1 is remarkable because it says that the event $\{(\hat{E}, \hat{s}) = (E, s)\}$ can be rewritten as affine constraints on $\mathbf{y}$. This is because $\mathbf{w}$ and $\mathbf{u}$ are already affine functions of $\mathbf{y}$, and the constraints $\text{sign}(\cdot) = \mathbf{s}$ and $\|\cdot\|_\infty < 1$ can also be rewritten in terms of affine constraints. The following proposition makes this explicit.

PROPOSITION 4.2. *Let $\mathbf{w}$ and $\mathbf{u}$ be defined as in (4.5) and (4.6). Then:*

$$(4.8) \quad \{\text{sign}(\mathbf{w}) = \mathbf{s}, \|\mathbf{u}\|_\infty < 1\} = \left\{ \begin{pmatrix} A_0(E, \mathbf{s}) \\ A_1(E, \mathbf{s}) \end{pmatrix} \mathbf{y} < \begin{pmatrix} \mathbf{b}_0(E, \mathbf{s}) \\ \mathbf{b}_1(E, \mathbf{s}) \end{pmatrix} \right\}$$

where $A_0, \mathbf{b}_0$ encode the “inactive” constraints $\{\|\mathbf{u}\|_\infty < 1\}$, and $A_1, \mathbf{b}_1$ encode the “active” constraints $\{\text{sign}(\mathbf{w}) = \mathbf{s}\}$. These matrices have the explicit forms:

$$\begin{aligned} A_0(E, \mathbf{s}) &= \frac{1}{\lambda} \begin{pmatrix} X_{-E}^T(I - P_E) \\ -X_{-E}^T(I - P_E) \end{pmatrix} & \mathbf{b}_0(E, \mathbf{s}) &= \begin{pmatrix} \mathbf{1} - X_{-E}^T(X_E^T)^+ \mathbf{s} \\ \mathbf{1} + X_{-E}^T(X_E^T)^+ \mathbf{s} \end{pmatrix} \\ A_1(E, \mathbf{s}) &= -\text{diag}(\mathbf{s})(X_E^T X_E)^{-1} X_E^T & \mathbf{b}_1(E, \mathbf{s}) &= -\lambda \text{diag}(\mathbf{s})(X_E^T X_E)^{-1} \mathbf{s} \end{aligned}$$

PROOF. First, substituting expression (4.5) for $\mathbf{w}$, we rewrite the “active” constraints as

$$\begin{aligned} \{\text{sign}(\mathbf{w}) = \mathbf{s}\} &= \{\text{diag}(\mathbf{s})\mathbf{w} > 0\} \\ &= \{\text{diag}(\mathbf{s})(X_E^T X_E)^{-1}(X_E^T \mathbf{y} - \lambda \mathbf{s}) > 0\} \\ &= \{A_1(E, \mathbf{s})\mathbf{y} < \mathbf{b}_1(E, \mathbf{s})\}. \end{aligned}$$

Next, substituting expression (4.6) for $\mathbf{u}$, we rewrite the “inactive” constraints as

$$\begin{aligned} \{\|\mathbf{u}\|_\infty < 1\} &= \left\{ -\mathbf{1} < X_{-E}^T(X_E^T)^+ \mathbf{s} + \frac{1}{\lambda} X_{-E}^T(I - P_E)\mathbf{y} < \mathbf{1} \right\} \\ &= \{A_0(E, \mathbf{s})\mathbf{y} < \mathbf{b}_0(E, \mathbf{s})\} \end{aligned}$$

$\square$

Combining Lemma 4.1 with Proposition 4.2, we obtain the following.

THEOREM 4.3. $\{(\hat{E}, \hat{s}) = (E, s)\} = \{A(E, \mathbf{s})\mathbf{y} \leq \mathbf{b}(E, \mathbf{s})\}.$

As a corollary, $\{\hat{E} = E\}$ is simply the union of the above sets over all possible sign patterns.

$$\text{COROLLARY 4.4. } \{\hat{E} = E\} = \bigcup_{\mathbf{s} \in \{-1, 1\}^{|E|}} \{A(E, \mathbf{s})\mathbf{y} \leq \mathbf{b}(E, \mathbf{s})\}.$$

5. Conditioning on Polytopes. In order to obtain inference conditional on the model, we need to understand the distribution of

$$\boldsymbol{\eta}_E^T \mathbf{y} \mid \{\hat{E} = E\}.$$

However, as we saw in the previous section, $\{\hat{E} = E\}$ is a union of polytopes, so it is easier to condition on both the model *and the signs*,

$$\boldsymbol{\eta}_E^T \mathbf{y} \mid \{\hat{E} = E, \hat{\mathbf{s}} = \mathbf{s}\},$$

where the conditioning event is a single polytope $\{A(E, \mathbf{s})\mathbf{y} \leq \mathbf{b}(E, \mathbf{s})\}$. Notice that inferences that are valid conditional on this finer event will also be valid conditional on $\{\hat{E} = E\}$. For example, if a confidence interval C_j^E for β_j^E has $(1 - \alpha)$ coverage conditional on the model and signs

$$\mathbb{P}(\beta_j^E \in C_j^E \mid \hat{E} = E, \hat{\mathbf{s}} = \mathbf{s}) \geq 1 - \alpha,$$

then it will also have $(1 - \alpha)$ coverage conditional only on the model:

$$\begin{aligned} \mathbb{P}(\beta_j^E \in C_j^E \mid \hat{E} = E) &= \sum_{\mathbf{s}} \mathbb{P}(\beta_j^E \in C_j^E \mid \hat{E} = E, \hat{\mathbf{s}} = \mathbf{s}) \mathbb{P}(\hat{\mathbf{s}} = \mathbf{s} \mid \hat{E} = E) \\ &\geq \sum_{\mathbf{s}} (1 - \alpha) \mathbb{P}(\hat{\mathbf{s}} = \mathbf{s} \mid \hat{E} = E) \\ &= 1 - \alpha. \end{aligned}$$

This section is divided into two subsections. First, we study how to condition on a single polytope; this will allow us to condition on $\{\hat{E} = E, \hat{\mathbf{s}} = \mathbf{s}\}$. Then, we look at how to extend the framework to condition on a union of polytopes, which will allow us to condition only on the model $\{\hat{E} = E\}$. The inferences obtained by conditioning on the model will in general be more “optimal” (i.e., narrower intervals, more powerful tests), at the price of more computation.

5.1. Conditioning on a Single Polytope. Suppose we observe $\mathbf{y} \sim N(\boldsymbol{\mu}, \Sigma)$, and $\boldsymbol{\eta} \in \mathbb{R}^n$ is some direction of interest. To understand the distribution of

$$(5.1) \quad \boldsymbol{\eta}^T \mathbf{y} \mid \{A\mathbf{y} \leq \mathbf{b}\},$$

we rewrite $\{A\mathbf{y} \leq \mathbf{b}\}$ in terms of $\boldsymbol{\eta}^T \mathbf{y}$ and a component $\mathbf{z}$ which is independent of $\boldsymbol{\eta}^T \mathbf{y}$. That component is

$$(5.2) \quad \mathbf{z} \equiv (I_n - \mathbf{c}\boldsymbol{\eta}^T)\mathbf{y},$$

where

$$(5.3) \quad \mathbf{c} = \Sigma\boldsymbol{\eta}(\boldsymbol{\eta}^T \Sigma \boldsymbol{\eta})^{-1}.$$

It is easy to verify that $\mathbf{z}$ is uncorrelated with, and hence independent of, $\boldsymbol{\eta}^T \mathbf{y}$. Although definition (5.2) may seem unmotivated, in the case where $\Sigma = \sigma^2 I_n$, $\mathbf{z}$ is simply the residual $(I_n - P_{\boldsymbol{\eta}})\mathbf{y}$ from projecting $\mathbf{y}$ onto $\boldsymbol{\eta}$.

We can now rewrite $\{A\mathbf{y} \leq \mathbf{b}\}$ in terms of $\boldsymbol{\eta}^T \mathbf{y}$ and $\mathbf{z}$.

LEMMA 5.1. *Let $\mathbf{z}$ be defined as in (5.2) and $\mathbf{c}$ as in (5.3). Then, the conditioning set can be rewritten as follows:*

$$\{A\mathbf{y} \leq \mathbf{b}\} = \{\mathcal{V}^-(\mathbf{z}) \leq \boldsymbol{\eta}^T \mathbf{y} \leq \mathcal{V}^+(\mathbf{z}), \mathcal{V}^0(\mathbf{z}) \geq 0\}$$

where

$$(5.4) \quad \mathcal{V}^-(\mathbf{z}) \equiv \max_{j:(A\mathbf{c})_j < 0} \frac{b_j - (A\mathbf{z})_j}{(A\mathbf{c})_j}$$

$$(5.5) \quad \mathcal{V}^+(\mathbf{z}) \equiv \min_{j:(A\mathbf{c})_j > 0} \frac{b_j - (A\mathbf{z})_j}{(A\mathbf{c})_j}$$

$$(5.6) \quad \mathcal{V}^0(\mathbf{z}) \equiv \min_{j:(A\mathbf{c})_j = 0} b_j - (A\mathbf{z})_j.$$

Note that $\mathcal{V}^-$, $\mathcal{V}^+$, and $\mathcal{V}^0$ refer to functions. Since they are functions of $\mathbf{z}$ only, (5.4)-(5.6) are independent of $\boldsymbol{\eta}^T \mathbf{y}$.

PROOF. The proof is essentially contained in Figure 1.

[The algebra I presented in my talk needs to go here.] □

Lemma 5.1 tells us that

$$(5.7) \quad [\boldsymbol{\eta}^T \mathbf{y} \mid \{A\mathbf{y} \leq \mathbf{b}\}] \stackrel{d}{=} [\boldsymbol{\eta}^T \mathbf{y} \mid \{\mathcal{V}^-(\mathbf{z}) \leq \boldsymbol{\eta}^T \mathbf{y} \leq \mathcal{V}^+(\mathbf{z}), \mathcal{V}^0(\mathbf{z}) \geq 0\}]$$

Since $\mathcal{V}^+(\mathbf{z})$, $\mathcal{V}^-(\mathbf{z})$, $\mathcal{V}^0(\mathbf{z})$ are independent of $\boldsymbol{\eta}^T \mathbf{y}$, they behave as “fixed” quantities. Thus, $\boldsymbol{\eta}^T \mathbf{y}$ is conditionally like a normal random variable, truncated to be between $\mathcal{V}^-(\mathbf{z})$ and $\mathcal{V}^+(\mathbf{z})$. We would like to be able to say

$$“\boldsymbol{\eta}^T \mathbf{y} \mid \{A\mathbf{y} \leq \mathbf{b}\} \sim TN(\boldsymbol{\eta}^T \boldsymbol{\mu}, \sigma^2 \boldsymbol{\eta}^T \Sigma \boldsymbol{\eta}, \mathcal{V}^-(\mathbf{z}), \mathcal{V}^+(\mathbf{z})),”$$

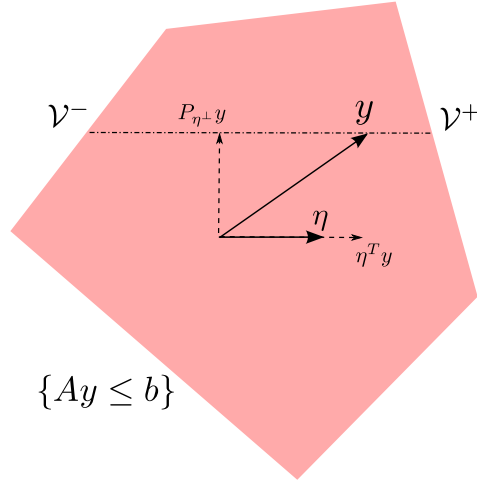


Fig 1: A picture demonstrating that the set $\{A\mathbf{y} \leq \mathbf{b}\}$ can be characterized by $\{\mathcal{V}^-(\mathbf{z}) \leq \boldsymbol{\eta}^T \mathbf{y} \leq \mathcal{V}^+(\mathbf{z})\}$. Assuming $\Sigma = I$ and $\|\boldsymbol{\eta}\|_2 = 1$, $\mathcal{V}^-(\mathbf{z})$ and $\mathcal{V}^+(\mathbf{z})$ are functions of $\mathbf{z}$ only, which is independent of $\boldsymbol{\eta}^T \mathbf{y}$.

but this is incorrect, since the distribution on the right-hand side changes with $\mathbf{z}$. To make the intuition precise, we use the probability integral transform to obtain a statistic $F^{\mathbf{z}}(\boldsymbol{\eta}^T \mathbf{y})$ that has a $\text{Unif}(0,1)$ distribution for any value of $\mathbf{z}$. Hence, $F^{\mathbf{z}}(\boldsymbol{\eta}^T \mathbf{y})$ will also have a $\text{Unif}(0,1)$ distribution marginally over $\mathbf{z}$. We make this precise in the next theorem.

THEOREM 5.2. *Let $F_{\mu, \sigma^2}^{[a,b]}$ denote the CDF of a $N(\mu, \sigma^2)$ random variable truncated to the interval $[a, b]$, i.e.:*

$$(5.8) \quad F_{\mu, \sigma^2}^{[a,b]}(x) = \frac{\Phi((x - \mu)/\sigma) - \Phi((a - \mu)/\sigma)}{\Phi((b - \mu)/\sigma) - \Phi((a - \mu)/\sigma)}$$

where Φ is the CDF of a $N(0,1)$ random variable. Then:

$$(5.9) \quad F_{\boldsymbol{\eta}^T \boldsymbol{\mu}, \boldsymbol{\eta}^T \Sigma \boldsymbol{\eta}}^{[\mathcal{V}^-(\mathbf{z}), \mathcal{V}^+(\mathbf{z})]}(\boldsymbol{\eta}^T \mathbf{y}) \mid \{A\mathbf{y} \leq \mathbf{b}\} \sim \text{Unif}(0,1)$$

where $\mathcal{V}^-$ and $\mathcal{V}^+$ are defined in (5.4) and (5.5).

PROOF. First, apply Lemma 5.1:

$$\begin{aligned} [\boldsymbol{\eta}^T \mathbf{y} \mid A\mathbf{y} \leq \mathbf{b}, \mathbf{z} = \mathbf{z}_0] &\stackrel{d}{=} [\boldsymbol{\eta}^T \mathbf{y} \mid \mathcal{V}^-(\mathbf{z}) \leq \boldsymbol{\eta}^T \mathbf{y} \leq \mathcal{V}^+(\mathbf{z}), \mathcal{V}^0(\mathbf{z}) \geq 0, \mathbf{z} = \mathbf{z}_0] \\ &\stackrel{d}{=} [\boldsymbol{\eta}^T \mathbf{y} \mid \mathcal{V}^-(\mathbf{z}_0) \leq \boldsymbol{\eta}^T \mathbf{y} \leq \mathcal{V}^+(\mathbf{z}_0), \mathcal{V}^0(\mathbf{z}_0) \geq 0, \mathbf{z} = \mathbf{z}_0] \end{aligned}$$

The only random quantities left are $\boldsymbol{\eta}^T \mathbf{y}$ and $\mathbf{z}$. Now we can eliminate $\mathbf{z} = \mathbf{z}_0$ from the condition using independence:

$$\begin{aligned} &\stackrel{d}{=} [\boldsymbol{\eta}^T \mathbf{y} | \mathcal{V}^-(\mathbf{z}_0) \leq \boldsymbol{\eta}^T \mathbf{y} \leq \mathcal{V}^+(\mathbf{z}_0)] \\ &\sim TN(\boldsymbol{\eta}^T \boldsymbol{\mu}, \sigma^2 \|\boldsymbol{\eta}\|^2, \mathcal{V}^-(\mathbf{z}_0), \mathcal{V}^+(\mathbf{z}_0)) \end{aligned}$$

Letting $F^{\mathbf{z}}(\boldsymbol{\eta}^T \mathbf{y}) \equiv F_{\boldsymbol{\eta}^T \boldsymbol{\mu}, \boldsymbol{\eta}^T \Sigma \boldsymbol{\eta}}^{[\mathcal{V}^-(\mathbf{z}), \mathcal{V}^+(\mathbf{z})]}(\boldsymbol{\eta}^T \mathbf{y})$, we can apply the probability integral transform to the above result to obtain

$$\begin{aligned} [F^{\mathbf{z}}(\boldsymbol{\eta}^T \mathbf{y}) | A\mathbf{y} \leq \mathbf{b}, \mathbf{z} = \mathbf{z}_0] &\stackrel{d}{=} [F^{\mathbf{z}_0}(\boldsymbol{\eta}^T \mathbf{y}) | A\mathbf{y} \leq \mathbf{b}, \mathbf{z} = \mathbf{z}_0] \\ &\sim \text{Unif}(0, 1) \end{aligned}$$

If we let p_X denote the density of a random variable X given $\{A\mathbf{y} \leq \mathbf{b}\}$, what we have just shown is that

$$p_{F^{\mathbf{z}}(\boldsymbol{\eta}^T \mathbf{y}) | \mathbf{z}}(f | \mathbf{z}_0) \equiv \frac{p_{F^{\mathbf{z}}(\boldsymbol{\eta}^T \mathbf{y}), \mathbf{z}}(f, \mathbf{z}_0)}{p_{\mathbf{z}}(\mathbf{z}_0)} = 1_{[0,1]}(f)$$

for any $\mathbf{z}_0$. The desired result now follows by integrating over $\mathbf{z}_0$:

$$\begin{aligned} p_{F^{\mathbf{z}}(\boldsymbol{\eta}^T \mathbf{y})}(f) &= \int p_{F^{\mathbf{z}}(\boldsymbol{\eta}^T \mathbf{y}) | \mathbf{z}}(f | \mathbf{z}_0) p_{\mathbf{z}}(\mathbf{z}_0) d\mathbf{z}_0 \\ &= \int 1_{[0,1]}(f) p_{\mathbf{z}}(\mathbf{z}_0) d\mathbf{z}_0 \\ &= 1_{[0,1]}(f). \end{aligned}$$

□

In Figure 2, we plot the density of the truncated Gaussian, noting that its shape depends on the location of μ relative to $[a, b]$ as well as the width relative to σ .

5.2. *Conditioning on a Union of Polytopes.* [Move the "minimal" interval section here.]

6. Application to Inference for the Lasso. In this section, we apply the theory developed in in Sections 4 and 5 to the lasso. In particular, we will construct confidence intervals for the active variables and test the chosen model based on the pivot developed in Section 5.

To summarize the developments so far, recall that our model says that $y \sim N(\mu, \sigma^2 I)$. The distribution of interest is $y \mid \{(\hat{E}, \hat{s}) = (E, s)\}$. By Theorem

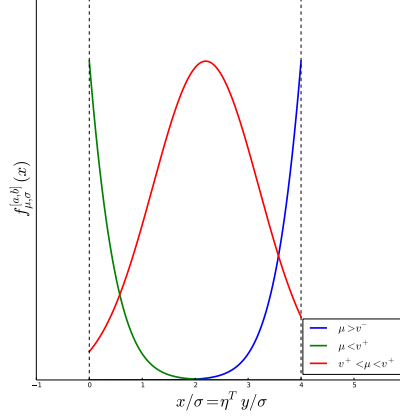


Fig 2: The density of the truncated Gaussian with distribution $F_{\mu, \sigma^2}^{[v^-, v^+]}$ depends on the width of $[v^-, v^+]$ relative to σ as well as the location of μ relative to $[v^-, v^+]$. When μ is firmly inside the interval, the distribution resembles a Gaussian. As μ varies drifts outside $[v^-, v^+]$, the density begins to converge to an exponential distribution with mean inversely proportional to the distance between μ and its projection onto $[v^-, v^+]$.

4.1, this is equivalent to $y \mid \{A(E, s)y \leq b(E, s)\}$ defined in Proposition 4.2. Now we can apply Theorem 5.2 to obtain the (conditional) pivot

$$(6.1) \quad F_{\eta^T \mu, \sigma^2 \|\eta\|_2^2}^{[\mathcal{V}^-, \mathcal{V}^+]}(\eta^T y) \mid \{(\hat{E}, \hat{s}) = (E, s)\} \sim \text{Unif}(0, 1)$$

for any η , where $\mathcal{V}^-$ and $\mathcal{V}^+$ are defined in (5.4) and (5.5). Note that $A(E, s)$ and $b(E, s)$ appear in this pivot through $\mathcal{V}^-$ and $\mathcal{V}^+$. This pivot will play a central role in all of the applications that follow.

6.1. *Confidence Intervals for the Active Variables.* In this section, we describe how to form confidence intervals for the components of $\beta_E^* = X_E^+ \mu$. If we choose

$$(6.2) \quad \eta_j = (X_E^T)^+ e_j,$$

then $\eta_j^T \mu = \beta_{\hat{E}, j}^*$, so the above framework provides a method for inference about the j^{th} variable in the model $\hat{E}$. Note that this reduces to inference about the true β_j^0 if $\hat{E} \supset S := \{j : \beta_j^0 \neq 0\}$, as discussed in Section 1.

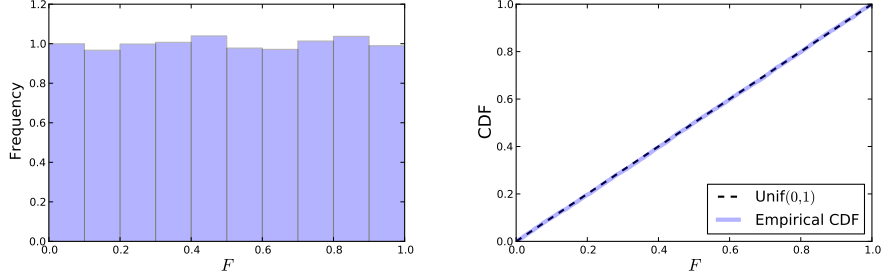


Fig 3: Histogram and empirical distribution of $F_{\eta^T \mu, \eta^T \Sigma \eta}^{[\mathcal{V}^-, \mathcal{V}^+]}(\eta^T y)$ obtained by sampling $y \sim N(\mu, \Sigma)$ constrained to $\{Ay \leq b\}$. The distribution is very close to $\text{Unif}(0, 1)$ as shown in Theorem 5.2.

The conditions under which this holds is well known in the literature, cf. Bühlmann and van de Geer (2011).

By applying Theorem 5.2, we obtain the following (conditional) pivot for $\beta_{\hat{E}, j}^*$:

$$F_{\beta_{\hat{E}, j}^*, \sigma^2 \|\eta_j\|^2}^{[\mathcal{V}^-, \mathcal{V}^+]}(\eta_j^T y) \mid \{(\hat{E}, \hat{s}) = (E, s)\} \sim \text{Unif}(0, 1).$$

Note that j and η_j are both random—but only through $\hat{E}$, a quantity which is fixed after conditioning—so Theorem 5.2 holds even for this “random” choice of η . The obvious way to obtain an interval is to “invert” the pivot. In other words, since

$$\mathbb{P} \left(\frac{\alpha}{2} \leq F_{\beta_{\hat{E}, j}^*, \sigma^2 \|\eta_j\|^2}^{[\mathcal{V}^-, \mathcal{V}^+]}(\eta_j^T y) \leq 1 - \frac{\alpha}{2} \mid \{(\hat{E}, \hat{s}) = (E, s)\} \right) = \alpha,$$

one can define a $(1 - \alpha)$ (conditional) confidence interval for $\beta_{\hat{E}, j}^*$ as

$$\left\{ \beta_{\hat{E}, j}^* : \frac{\alpha}{2} \leq F_{\beta_{\hat{E}, j}^*, \sigma^2 \|\eta_j\|^2}^{[\mathcal{V}^-, \mathcal{V}^+]}(\eta_j^T y) \leq 1 - \frac{\alpha}{2} \right\}.$$

In fact, F is monotone decreasing in $\beta_{\hat{E}, j}^*$, so to find its endpoints, one need only solve for the root of a smooth one-dimensional function. The monotonicity is a consequence of the fact that the truncated Gaussian distribution is a natural exponential family and hence has monotone likelihood ratio in μ . The details can be found in Appendix A.

We now formalize the above observations in the following result, an immediate consequence of Theorem 5.2.

COROLLARY 6.1. Let η_j be defined as in (6.2), and let $L_\alpha^j = L_\alpha^j(\eta_j, \hat{E}, \hat{s})$ and $U_\alpha^j = U_\alpha^j(\eta_j, \hat{E}, \hat{s})$ be the (unique) values satisfying

$$F_{L_\alpha^j, \sigma^2 \|\eta_j\|^2}^{[\mathcal{V}^-, \mathcal{V}^+]}(\eta_j^T y) = 1 - \alpha \quad F_{U_\alpha^j, \sigma^2 \|\eta_j\|^2}^{[\mathcal{V}^-, \mathcal{V}^+]}(\eta_j^T y) = \alpha$$

Then $[L_\alpha^j, U_\alpha^j]$ is a $(1 - \alpha)$ confidence interval for $\eta_j^T \mu$, conditional on $(\hat{E}, \hat{s})$:

$$(6.3) \quad \mathbb{P}\left(\eta_j^T \mu \in [L_\alpha^j, U_\alpha^j] \mid \{(\hat{E}, \hat{s}) = (E, s)\}\right) = 1 - \alpha.$$

If $\eta_j^T y$ are not near the boundaries $[\mathcal{V}^-, \mathcal{V}^+]$, then the intervals will be relatively short. This is shown in Figure 4. Figure 5 shows two simulations that demonstrate our intervals cover at the nominal rate. We leave an exhaustive study of such intervals for the lasso to future work, noting that the truncation framework described can be used to form intervals with exact coverage properties.

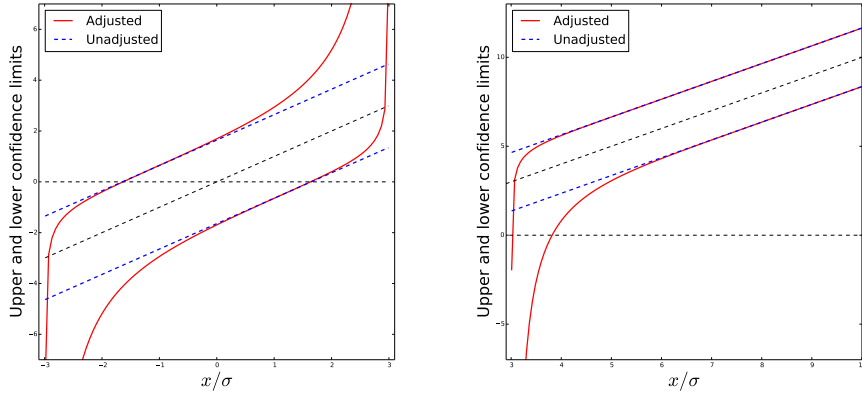


Fig 4: Upper and lower bounds of 90% confidence intervals based on $[a, b] = [-3\sigma, 3\sigma]$ as a function of the observation x/σ . We see that as long as the observation x/σ is roughly 0.5σ away from either boundary, the size of the intervals is comparable to an unadjusted confidence interval.

6.2. *Testing the Lasso-Selected Model.* Having observed that the lasso selected the variables $\hat{E}$, another relevant question is whether it has captured all of the signal in the model, i.e.,

$$(6.4) \quad H_0 : \beta_{-\hat{E}}^0 = \mathbf{0}.$$

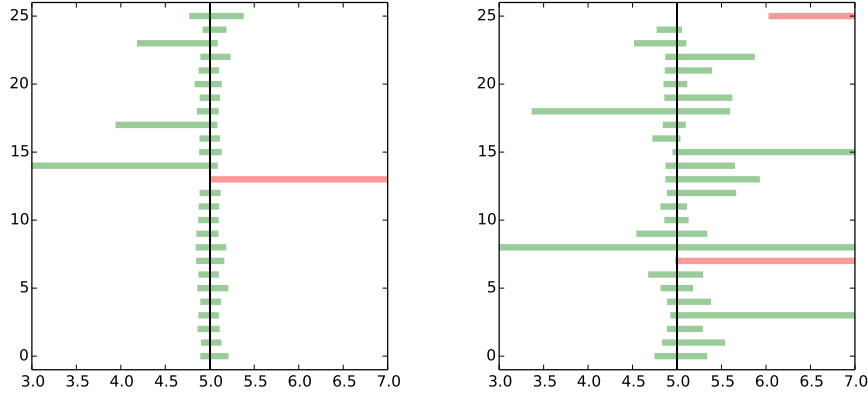


Fig 5: 90% confidence intervals for $\eta_1^T \mu$ for a small ($n = 100$, $p = 50$) and a large ($n = 100$, $p = 200$) uncorrelated Gaussian design, computed over 25 simulated data sets. The true model has five non-zero coefficients, all set to 5.0, and the noise variance is 0.25. A green bar means the confidence interval covers the true value while a red bar means otherwise.

We consider a slightly more general question, which does not assume the correctness of the linear model $\mu = X\beta^0$ and also takes into account whether the non-selected variables can improve the fit:

$$(6.5) \quad H_0 : X_{-\hat{E}}^T (I - P_{\hat{E}}) \mu = \mathbf{0}.$$

This quantity is the partial correlation of the non-selected variables with μ , adjusting for the variables in $\hat{E}$. This is more general because if we assume $\mu = X\beta^0$ for some β^0 and X is full rank, then rejecting (6.5) implies that there exists $i \in \text{supp}(\beta^0)$ not in $\hat{E}$, so we would also reject (6.4).

The natural approach is to compare the observed partial correlations $X_{-E}^T (I - P_E) y$ to $\mathbf{0}$. However, the framework of Section 5 only allows tests of μ in a single direction η . To make use of that framework, we can choose η such that it selects the maximum magnitude of $X_{-E}^T (I - P_E) y$. In particular, this direction provides the most evidence against the null hypothesis of zero partial correlation, so if the null hypothesis cannot be rejected in this direction, it would not be rejected in any direction.

Letting $j^* := \arg\max_j |e_j^T X_{-E}^T (I - P_E) y|$ and $s_j := \text{sign}(e_j^T X_{-E}^T (I - P_E) y)$, we set

$$(6.6) \quad \eta_{j^*} = s_{j^*} (I - P_E) X_{-E} e_{j^*},$$

and test $H_0 : \eta_{j^*}^T \mu = 0$. However, the results in Section 5 cannot be directly applied to this setting because j^* and s_{j^*} are random variables that are not measurable with respect to $(\hat{E}, \hat{s})$.

To resolve this issue, we propose a test conditional not only on $(\hat{E}, \hat{s})$, but also on the index and sign of the maximizer:

$$(6.7) \quad \{(\hat{E}, \hat{s}) = (E, s), (j^*, s_{j^*}) = (j, s)\}.$$

A test that is level α conditional on (6.7) for all (E, s) and (j, s) is also level α conditional on $\{(\hat{E}, \hat{s}) = (E, s)\}$.

In order to use the results of Section 5, we must show that (6.7) can be written in the form $A(E, s, j, s)y \leq b(E, s, j, s)$. This is indeed possible, and the following proposition provides an explicit construction.

PROPOSITION 6.2. *Let A_0, b_0, A_1, b_1 be defined as in Proposition 4.2. Then:*

$$\{(\hat{E}, \hat{s}) = (E, s), (j^*, s_{j^*}) = (j, s)\} = \left\{ \begin{pmatrix} A_0(E, s) \\ A_1(E, s) \\ A_2(E, j, s) \end{pmatrix} y < \begin{pmatrix} b_0(E, s) \\ b_1(E, s) \\ \mathbf{0} \end{pmatrix} \right\}$$

where $A_2(E, j, s)$ is defined as

$$A_2(E, j, s) = -s \begin{pmatrix} D_j(E) \\ S_j(E) \end{pmatrix} X_{-E}^T (I - P_E)$$

and D_j and S_j are $(|E| - 1) \times |E|$ operators that compute the difference and sum, respectively, of the j^{th} element with the other elements, e.g.,

$$D_1 = \begin{pmatrix} 1 & -1 & & & \\ 1 & & -1 & & \\ & & & \ddots & \\ & & & & -1 \\ 1 & & & & \end{pmatrix} \quad S_1 = \begin{pmatrix} 1 & 1 & & & \\ 1 & & 1 & & \\ & & & \ddots & \\ & & & & 1 \end{pmatrix}.$$

PROOF. The constraints $\{A_0 y < b_0\}$ and $\{A_1 y < b_1\}$ come from Proposition (4.2) and encode the constraints $\{(\hat{E}, \hat{s}) = (E, s)\}$. We show that the last two sets of constraints encode $\{(j^*, s_{j^*}) = (j, s)\}$.

Let $r := X_{-E}^T (I - P_E) y$ denote the vector of partial correlations. If $s = +1$, then $|r_j| > |r_i|$ for all $i \neq j$ if and only if $r_j - r_i > 0$ and $r_j + r_i > 0$ for all $i \neq j$. We can write this as $D_j r > 0$ and $S_j r > 0$. If $s = -1$, then the signs are flipped: $D_j r < 0$ and $S_j r < 0$. This establishes

$$\{(j^*, s_{j^*}) = (j, s)\} = \left\{ -s \begin{pmatrix} D_j \\ S_j \end{pmatrix} r < \mathbf{0} \right\} = \{A_2 y < \mathbf{0}\}.$$

□

Because of Proposition 6.2, we can now obtain the following result as a simple consequence of Theorem 5.2, which says that $F_{0,\sigma^2\|\eta_{j^*}\|^2}^{[\mathcal{V}^-, \mathcal{V}^+]}(\eta_{j^*}^T y) \sim \text{Unif}(0, 1)$, conditional on the set (6.7) and H_0 . We reject when $F_{0,\sigma^2\|\eta_{j^*}\|^2}^{[\mathcal{V}^-, \mathcal{V}^+]}(\eta_{j^*}^T y)$ is large because $F_{0,\sigma^2\|\eta_{j^*}\|^2}^{[\mathcal{V}^-, \mathcal{V}^+]}(\cdot)$ is monotone increasing in the argument and $\eta_{j^*}^T \mu$ is likely to be positive under the alternative.

COROLLARY 6.3. *Let H_0 and η_{j^*} be defined as in (6.6). Then, the test which rejects when*

$$\left\{ F_{0,\sigma^2\|\eta_{j^*}\|^2}^{[\mathcal{V}^-, \mathcal{V}^+]}(\eta_{j^*}^T y) > 1 - \alpha \right\}$$

is level α , conditional on $\{(\hat{E}, \hat{s}) = (E, s), (j^, s_{j^*}) = (j, s)\}$. That is,*

$$\mathbb{P} \left(F_{0,\sigma^2\|\eta_{j^*}\|^2}^{[\mathcal{V}^-, \mathcal{V}^+]}(\eta_{j^*}^T y) > 1 - \alpha \mid \{(\hat{E}, \hat{s}) = (E, s), (j^*, s_{j^*}) = (j, s)\} \cap H_0 \right) = \alpha.$$

In particular, since this holds for every (E, s, j, s) , this test also controls Type I error conditional only on $(\hat{E}, \hat{s})$, and unconditionally:

$$\begin{aligned} \mathbb{P} \left(F_{0,\sigma^2\|\eta_{j^*}\|^2}^{[\mathcal{V}^-, \mathcal{V}^+]}(\eta_{j^*}^T y) > 1 - \alpha \mid \{(\hat{E}, \hat{s}) = (E, s)\} \cap H_0 \right) &= \alpha \\ \mathbb{P} \left(F_{0,\sigma^2\|\eta_{j^*}\|^2}^{[\mathcal{V}^-, \mathcal{V}^+]}(\eta_{j^*}^T y) > 1 - \alpha \mid H_0 \right) &= \alpha. \end{aligned}$$

Figures 6 and 7 show the results of four simulation studies that demonstrate that the p-values are uniformly distributed when $H_{0,\lambda}$ is true and stochastically smaller than $\text{Unif}(0, 1)$ when it is false.

6.3. False Discovery Rate. False discovery rate (FDR) is defined as,

$$\mathbf{E} \left[\frac{V}{R} \right],$$

where V is the number of incorrectly rejected hypotheses and R is the total number of rejected hypotheses. We will restrict ourselves to the case of the well-specified linear model, $y = X\beta^0 + \epsilon$, and $n \geq p$ with X having full rank. In the context of linear regression, there is a sequence of hypotheses $H_{0,j} : \beta_j^0 = 0$ and a hypothesis is considered to be incorrectly rejected if $H_{0,j}$ is true, yet the variable is selected.

For each coefficient $j \in \hat{E}$, Equation (6.1) provides a valid p-value,

$$(6.8) \quad p_j = F_{\beta_j^0, \sigma^2\|\eta_j\|_2^2}^{[\mathcal{V}^-, \mathcal{V}^+]}(\eta_j^T y),$$

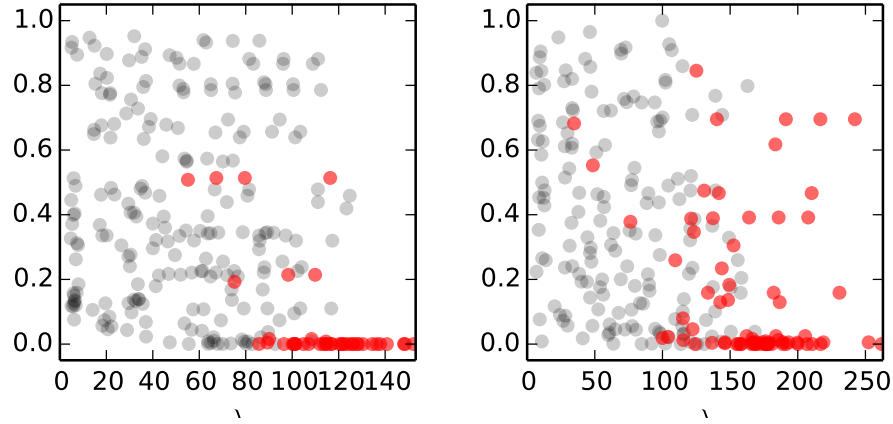


Fig 6: P-values for $H_{0,\lambda}$ at various λ values for a small ($n = 100$, $p = 50$) and a large ($n = 100$, $p = 200$) uncorrelated Gaussian design, computed over 50 simulated data sets. The true model has three non-zero coefficients, all set to 1.0, and the noise variance is 2.0. We see the p-values are Unif(0, 1) when the selected model includes the truly relevant predictors (black dots) and are stochastically smaller than Unif(0, 1) when the selected model omits a relevant predictor (red dots).

where $\eta_j = (X^T)^+ e_j$. For a null j , we have previously shown that $p_j | \{j \in \hat{E}\} \sim \text{Unif}(0, 1)$. Given p-values, we can now apply the Benjamini-Yekutieli procedure (Benjamini et al., 2001) for FDR control. Let $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(|\hat{E}|)}$ be the order statistics, and $h_{|\hat{E}|} = \sum_{i=1}^{|\hat{E}|} \frac{1}{i}$. Let k be

$$(6.9) \quad k = \max \left\{ k : p_{(k)} \leq \frac{k}{|\hat{E}| h_{|\hat{E}|}} \alpha \right\},$$

then reject $p_{(1)}, \dots, p_{(k)}$.

THEOREM 6.4. *Consider the procedure that forms p-values using Equation (6.8), chooses k via Equation (6.9), and rejects $p_{(1)}, \dots, p_{(k)}$. Then FDR is controlled at level α .*

PROOF. Conditioned on the event that variable j is in the lasso equicorrelation set, $j \in \hat{E}$, then p_j is uniformly distributed among the null variables. Applying the Benjamini-Yekutieli procedure to the p-values $p_{(1)}, \dots, p_{(|\hat{E}|)}$ guarantees FDR, since the null p-values are uniformly distributed. $\square$

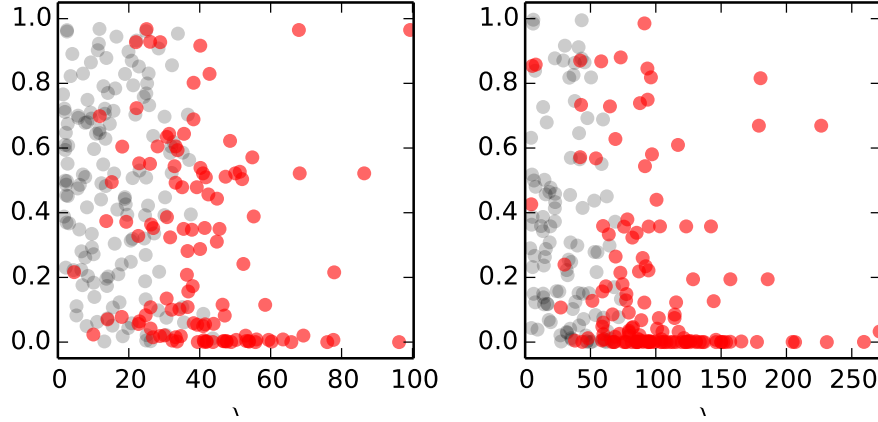


Fig 7: P-values for $H_{0,\lambda}$ at various λ values for a small ($n = 100$, $p = 50$) and a large ($n = 100$, $p = 200$) *correlated* ($\rho = 0.7$) Gaussian design, computed over 50 simulated data sets. The true model has three non-zero coefficients, all set to 1.0, and the noise variance is 2.0. Since the predictors are correlated, the relevant predictors are not always selected first. However, the p-values remain uniformly distributed when $H_{0,\lambda}$ is true and stochastically smaller than $\text{Unif}(0, 1)$ otherwise.

REMARK 6.5. *It is crucial that the p-values were formed conditional on $j \in \hat{E}$. If standard z-test p-values were used, the Benjamini Yekutieli procedure would have to be applied with $k = \max\{k : p_{(k)} \leq \frac{k}{p h_p} \alpha\}$. This procedure would be very conservative since $|\hat{E}| \ll p$.*

7. Data Example. We illustrate the application of inference for the lasso to the diabetes data set from [Efron et al. \(2004\)](#). First, all variables were standardized. Then, we chose λ according to the strategy in [Negahban et al. \(2012\)](#), $\lambda = 2 \mathbf{E}(\|X^T \epsilon\|_\infty)$, using an estimate of σ from the full model, resulting in $\lambda \approx 190$. The lasso selected four variables: BMI, BP, S3, and S5.

The intervals are shown in Figure 8, alongside the unadjusted confidence intervals produced by fitting OLS to the four selected variables, ignoring the selection. The latter is not a valid confidence interval conditional on the model. Also depicted are the confidence intervals obtained by *data splitting*; that is, if one splits the n observations into two halves, then uses one half for model selection and the other for inference. This is a competitor method that also produces valid confidence intervals conditional on the model. In this case, data splitting selected the same four variables, and the confidence

intervals were formed based on OLS on the half of the data set not used for model selection.

We can make two main observations from Figure 8.

1. The adjusted intervals provided by our method essentially reproduces the OLS intervals for the strong effects, whereas data splitting results in a loss of power by roughly a factor of $\sqrt{2}$ (since only $n/2$ observations are used in the inference).
2. One variable, **S3**, which would have been deemed significant using the OLS intervals, is no longer significant after adjustment. This demonstrates that taking model selection into account can have substantive impacts on the conclusions that are made.

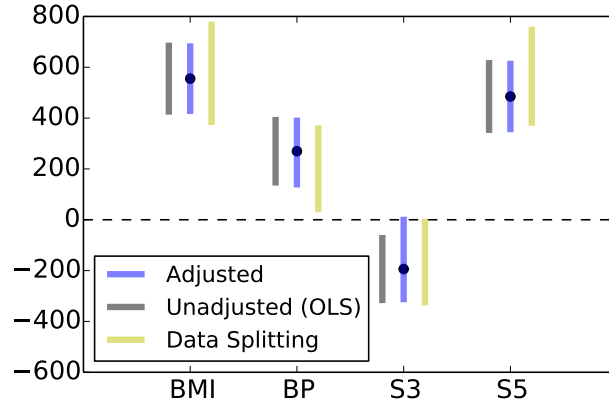


Fig 8: Inference for the four variables selected by the lasso ($\lambda = 190$) on the diabetes data set. The point estimate and adjusted confidence intervals using the approach in Section 6 are shown in blue. The gray show the OLS intervals, which ignore selection. The yellow lines show the intervals produced by splitting the data into two halves, forming the interval based on only half of the data.

8. Minimal Post-Selection Inference. We have described how to perform post-selection inference for the lasso conditional on both the equicorrelation set and signs $\{(\hat{E}, \hat{z}_{\hat{E}}) = (E, s)\}$. However, recall from Section 1 that the goal was inference conditional solely on the model, i.e., $\{\hat{E} = E\}$. In this section, we extend our framework to this setting, which we call minimal post-selection inference because we condition on the minimal set necessary

for the random η to be measurable. This results in more precise confidence intervals at the expense of greater computational cost.

To this end, we note that $\{\hat{E} = E\}$ is simply

$$\bigcup_{s \in \{-1, 1\}^{|E|}} \{(\hat{E}, \hat{z}_{\hat{E}}) = (E, s)\},$$

where the union is taken over all choices of signs. Therefore, the distribution of y conditioned on only the equicorrelation set $\{\hat{E} = E\}$ is a Gaussian vector constrained to a union of polytopes

$$y \mid \bigcup_{s \in \{-1, 1\}^{|E|}} \{A(E, s)y \leq b(E, s)\},$$

where $A(E, s)$ and $b(E, s)$ are given by (4.2).

To obtain inference about $\eta^T \mu$, we follow the arguments in Section 5 to obtain that this conditional distribution is equivalent to

$$(8.1) \quad \eta^T y \mid \bigcup_{s \in \{-1, 1\}^{|E|}} \{\mathcal{V}_s^-(y) \leq \eta^T y \leq \mathcal{V}_s^+(y), \mathcal{V}_s^0(y) \geq 0\},$$

where $\mathcal{V}_{z_E}^-$, $\mathcal{V}_{z_E}^+$, $\mathcal{V}_{z_E}^0$ are defined according to (5.4), (5.5), (5.6) with $A = A(E, s)$ and $b = b(E, z_E)$. Moreover, all of these quantities are still independent of $\eta^T y$, so instead of having a Gaussian truncated to a single interval $[\mathcal{V}^-, \mathcal{V}^+]$ as in Section 5, we now have a Gaussian truncated to the union of intervals $\bigcup_s [\mathcal{V}_s^-, \mathcal{V}_s^+]$. The geometric intuition is illustrated in Figure 9.

Finally, the probability integral transform once again yields a pivot:

$$F_{\eta^T \mu, \eta^T \Sigma \eta}^{\bigcup_{z_E} [\mathcal{V}_{z_E}^-(y), \mathcal{V}_{z_E}^+(y)]}(\eta^T y) \mid \{\hat{E} = E\} \sim \text{Unif}(0, 1).$$

It is now more useful to think of the notation of F as indicating the truncation set $C \subset \mathbb{R}$:

$$(8.2) \quad F_{\mu, \sigma^2}^C(x) := \frac{\Phi((-\infty, x] \cap C)}{\Phi(C)},$$

where Φ is the law of a $N(0, 1)$ random variable. We summarize these results in the following theorem.

THEOREM 8.1. *Let $F_{\mu, \sigma^2}^{\bigcup_i [a_i, b_i]}$ be the CDF of a normal truncated to the union of intervals $\bigcup_i [a_i, b_i]$, i.e., given by (8.2). Then:*

$$(8.3) \quad F_{\eta^T \mu, \eta^T \Sigma \eta}^{\bigcup_s [\mathcal{V}_s^-(y), \mathcal{V}_s^+(y)]}(\eta^T y) \mid \{\hat{E} = E\} \sim \text{Unif}(0, 1),$$

where $\mathcal{V}_s^-(y)$ and $\mathcal{V}_s^+(y)$ are defined in (5.4) and (5.5) with $A = A(E, s)$ and $b = b(E, s)$.

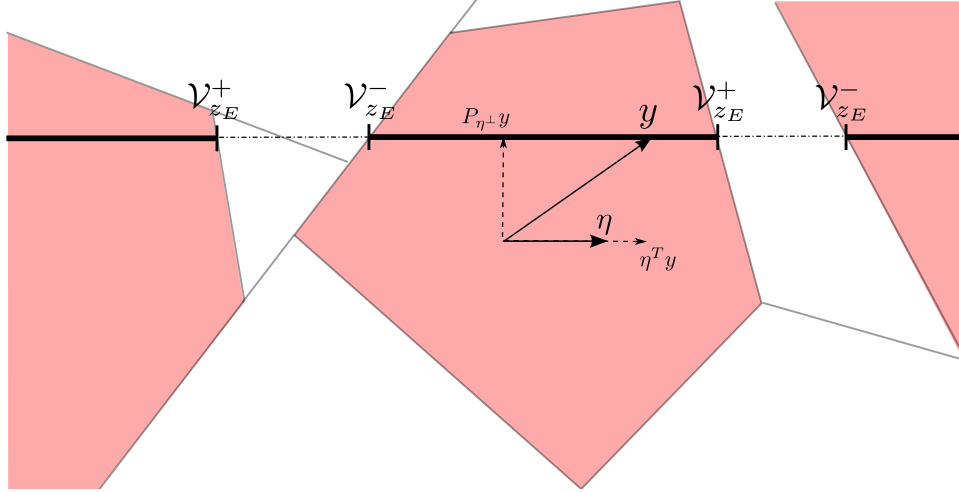


Fig 9: A picture demonstrating the effect of taking a union over signs. The polytope in the middle corresponds to the $(\hat{E}, \hat{s})$ that was observed and is the same polytope as in Figure 1. The difference is that we now consider potential (E, s) in addition to the one that was observed. The polytopes for the other (E, s) which have the same equicorrelation set $\hat{E}$ are red. The conditioning set is the union of these polytopes. We see that for y to be in this union, $\eta^T y$ must be in $\bigcup_s [\mathcal{V}_s^-, \mathcal{V}_s^+]$. The key point is that all of the $\mathcal{V}_s^-$ and $\mathcal{V}_s^+$ are still functions of only $P_{\eta^\perp} y$ and so are independent of $\eta^T y$.

The derivations of the confidence intervals and hypothesis tests in Section 6 remain valid using (8.3) as the pivot instead of (6.1). Figure 10 illustrates the effect of minimal post-selection inference in a simulation study, as compared with the “simple” inference described previously. The intervals are similar in most cases, but one can obtain great gains in precision using the minimal intervals when the simple intervals are very wide.

However, the tradeoff for this increased precision is greater computational cost. We computed $\mathcal{V}_s^-$ and $\mathcal{V}_s^+$ for all $s \in \{-1, 1\}^{|E|}$, which is only feasible when $|E|$ is fairly small. In what follows, we revert to the simple intervals described in Section 6, but extensions to the minimal inference setting are straightforward.

9. Extensions.

9.1. *Intervals for coefficients in full model.* Assuming the linear model $\mu = X\beta^0$ is correct, one might be interested in confidence intervals for β_j^0 instead of $\beta_{\hat{E},j}^*$. In this case, the target is not selection-dependent, so when

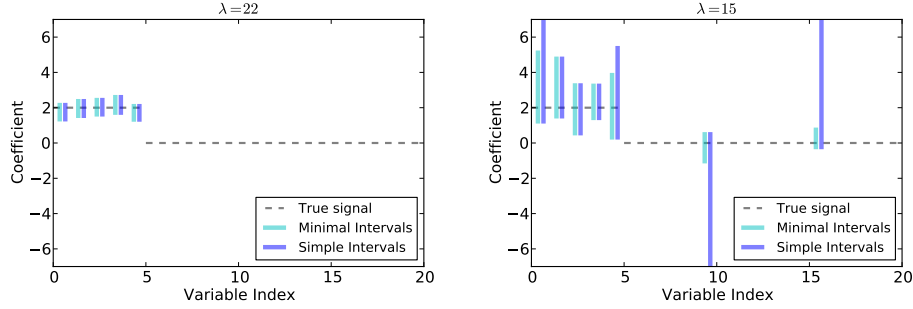


Fig 10: Comparison of the minimal and simple intervals as applied to the same simulated data set for two values of λ . The simulated data featured $n = 25$, $p = 50$, and 5 true non-zero coefficients; only the first 20 coefficients are shown. (We have included variables with no intervals to emphasize that inference is only on the selected variables.) We see that the simple intervals are virtually as good as the minimal intervals most of the time; the advantage of the minimal intervals is realized when the estimate is unstable and the simple intervals are very long, as in the right plot.

$p < n$, one can simply fit least squares and the standard OLS intervals will be valid (provided that one has a reasonable estimate of σ^2 when $p \approx n$). However, this approach is not possible when $p > n$.

One proposal that has been suggested by numerous authors ([Javanmard and Montanari, 2013](#); [van de Geer et al., 2013](#); [Zhang and Zhang, 2014](#)) is to use a debiased version of the lasso estimator:

$$\hat{\beta}^d = \hat{\beta} + \Theta X^T (y - X \hat{\beta})$$

where $\hat{\beta}$ denotes the lasso estimate and Θ is an approximate inverse of $X^T X$. From the representation

$$(9.1) \quad \hat{\beta}^d = \beta^0 + \Theta X^T \epsilon + \underbrace{(I - \Theta X^T X)(\hat{\beta} - \beta_0)}_{\hat{\Delta}}, \quad \epsilon \sim N(0, \Sigma),$$

the authors provide sufficient conditions for $\hat{\Delta} \rightarrow 0$, thereby making $\hat{\beta}^d$ asymptotically unbiased for β_0 .

We now show that in our framework, $\hat{\beta}^d$ is estimating a selection dependent target $\beta^d = \beta^d(\hat{E}, \hat{s})$. By now familiar manipulations of the optimality conditions (4.2),

$$\hat{\beta}_{\hat{E}} = X_{\hat{E}}^+ y - \lambda (X_{\hat{E}}^T X_{\hat{E}})^{-1} \hat{z}_{\hat{E}}.$$

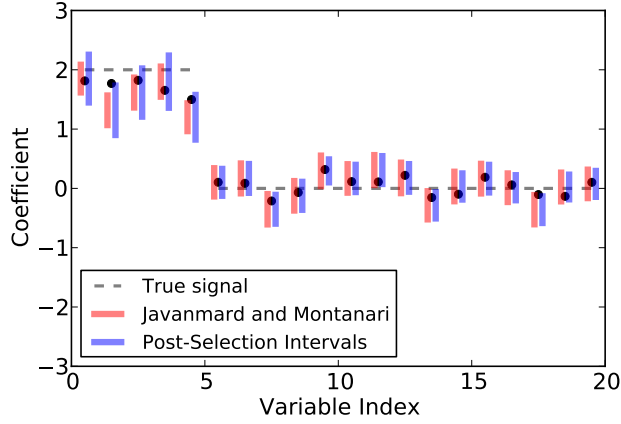


Fig 11: Confidence intervals for the coefficients in a design with $n = 25$, $p = 50$, and 5 non-zero coefficients. Only the first 20 coefficients are shown. The dotted line represents the true signal, and the points represent the (biased) post-selection target. The colored bars denote the intervals.

Assuming that $X = [X_{\hat{E}} : X_{-\hat{E}}]$, this allows us to write

$$(9.2) \quad \hat{\beta}^d = (I - \Theta X^T X) \begin{pmatrix} X_{\hat{E}}^+ y - \lambda(X_{\hat{E}}^T X_{\hat{E}})^{-1} \hat{s} \\ \mathbf{0} \end{pmatrix} + \Theta X^T y$$

On the set $\{(\hat{E}, \hat{z}_{\hat{E}}) = (E, z_E)\}$, $\hat{\beta}^d$ is simply an affine function of y . Thus, we can write $\hat{\beta}_j^d$ in the form $\eta_j^T y + h_j$, which can be thought of in the post-selection framework as estimating $\beta_j^d := \eta_j^T \mu + h_j$. If $\mu = X\beta^0$, then one obtains

$$\beta^d = \beta^0 + \underbrace{(I - \Theta X^T X) \begin{pmatrix} (X_{\hat{E}}^+ \mu - \lambda(X_{\hat{E}}^T X_{\hat{E}})^{-1} \hat{s}) \\ \mathbf{0} \end{pmatrix}}_{\Delta} - \beta^0.$$

Δ can be thought of as the population analog (conditional on $\hat{E}, \hat{s}$) of $\hat{\Delta}$ in (9.1). The post-selection framework of Section 5 allows us to construct exact conditional intervals for $\beta_j^d = \beta_j^0 + \Delta_j$. Under the assumptions about the choice of Θ and high-dimensional consistency in Javanmard and Montanari (2013), $\Delta_j \rightarrow 0$, so β_j^d is equivalent to β_j^0 .

Figure 11 shows the results of a simulation study. It makes clear that the intervals of Javanmard and Montanari (2013) do not necessarily cover the truth, but rather a biased target. Our post-selection intervals also cover the same biased target conditional on $(\hat{E}, \hat{s})$.

9.2. *Elastic net.* One problem with the lasso is that it tends to select only one variable out of a set of correlated variables, resulting in estimates which are unstable. The elastic net (Zou and Hastie, 2005) adds an ℓ_2 penalty to the lasso objective in order to stabilize the estimates:

$$(9.3) \quad \hat{\beta}^e = \operatorname{argmin}_{\beta} \frac{1}{2} \|y - X\beta\|_2^2 + \lambda \|\beta\|_1 + \frac{\gamma}{2} \|\beta\|_2^2.$$

Using a nearly identical argument to the one in Section 4, we see that necessary and sufficient conditions for $\{(\hat{E}, \hat{s}) = (E, s)\}$ are the existence of $U(E, s)$ and $W(E, s)$ satisfying

$$\begin{aligned} (X_E^T X_E + \gamma I)U - X_E^T y + \lambda s &= 0 \\ X_{-E}^T X_E U - X_{-E}^T y + \lambda W &= 0 \\ \operatorname{sign}(U) &= s, \quad W \in (-1, 1). \end{aligned}$$

Solving for U and W , we see that the selection event can be written

$$(9.4) \quad \{(\hat{E}, \hat{s}) = (E, s)\} = \left\{ \begin{pmatrix} A_0(E, s) \\ A_1(E, s) \end{pmatrix} y < \begin{pmatrix} b_0(E, s) \\ b_1(E, s) \end{pmatrix} \right\}$$

where A_0 , A_1 , b_0 , and b_1 are the same as in Proposition 4.2, except replacing $(X_E^T X_E)^{-1}$, which appears in the expressions through P_E and $(X_E^T)^+$, by the “damped” version $(X_E^T X_E + \gamma I)^{-1}$.

Having rewritten the selection event in the form (9.4), we can once again apply the framework of Section 5 to obtain a test for the elastic net conditional on this event.

9.3. *Alternative norms as test statistics.* In Section 6.2 we used the test statistic

$$T_\infty = \|X_{-\hat{E}}^T (I - P_{\hat{E}})y\|_\infty$$

and its conditional distribution on $\{(\hat{E}, \hat{s}) = (E, s)\}$ to test whether we had missed any large partial correlations in using $\hat{E}$ as the estimated active set. If we have indeed missed some variables in E there is no reason to suppose that the mean of $X_{-E}^T (I - P_E)y$ is sparse; hence the ℓ_∞ norm may not be the best norm to use as a test statistic.

In principle, we could have used virtually any norm, as long as we can say something about the distribution of this norm conditional on $\{(\hat{E}, \hat{s}) = (E, s)\}$. Problems of this form are considered in Taylor et al. (2013). For example, if we consider the quadratic

$$T_2 = \|X_{-E}^T (I - P_E)y\|_2$$

the general approach in [Taylor et al. \(2013\)](#) derives the conditional distribution of T_2 conditioned on

$$\eta_2^* = \arg \max_{\|\eta\|_2 \leq 1} \eta^T (X_{-E}^T (I - P_E) y).$$

In general, this distribution will be a χ^2 subject to random truncation as in Section 5 (see the group lasso examples in [Taylor et al. \(2013\)](#)). Adding the constraints encoded by $\{(\hat{E}, \hat{s}) = (E, s)\}$ affects only the random truncation $[\mathcal{V}^-, \mathcal{V}^+]$.

9.4. *Estimation of σ^2 .* As noted above, all of our results rely on a reliable estimate of σ^2 . While there are several approaches to estimating σ^2 in the literature, the truncated Gaussian theory described in this work itself provides a natural estimate.

Suppose the linear model is correct ($\mu = X\beta^0$). Then, on the event $\{\hat{E} = E, \hat{E} \supset S\}$, which we assume, the residual

$$(I - P_E)y$$

is a (multivariate) truncated Gaussian with mean $\mathbf{0}$, with law

$$\mathbb{P}_{C, \sigma^2}(B) = \mathbb{P}(Z \in B | Z \in C), \quad Z \sim N(\mathbf{0}, \sigma^2 I).$$

As σ^2 varies, one obtains a one-parameter exponential family with density

$$\frac{d\mathbb{P}_{C, \sigma^2}}{dz} = e^{-\alpha \|z\|_2^2 - \Lambda_C(\alpha)} 1_C(z)$$

and natural parameter $\alpha = \sigma^2/2$. On the event $\{(\hat{E}, \hat{s}) = (E, s)\}$, we set

$$C = \{y : A(E, z_E)y \leq b(E, z_E)\},$$

and then choose α (or equivalently, σ^2) to satisfy the score equation

$$(9.5) \quad \mathbb{E}_{C, \hat{\sigma}^2}(\|Z\|_2^2) = \|(I - P_E)y\|_2^2.$$

This amounts to a maximum likelihood estimate of σ^2 . The expectation on the left is generally impossible to do analytically, but there exist fast algorithms for sampling from $\mathbb{P}_{C, \sigma^2}$, c.f. [Geweke \(1991\)](#); [Rodriguez-Yam et al. \(2004\)](#). A rough outline of a naive version of such algorithms is to pick a direction such as e_i one of the coordinate axes. Based on the current state of Z , draw a new entry for the Z_i from the appropriate univariate truncated normal determined from the cutoffs described in Section 5. We repeat this procedure to evaluate the expectation on the left, and use gradient descent to find $\hat{\sigma}^2$.

9.5. *Composite Null Hypotheses.* In Section 6, we considered hypotheses of the form $H_0 : \eta_{j^\star}^T \mu = 0$, which said that the partial correlation of the variables in $-E$ with y , adjusting for the variables in E , was exactly 0. This may be unrealistic, and in practice, we may want to allow some tolerance for the partial correlation.

We consider testing instead the *composite* hypothesis

$$(9.6) \quad H_0 : |\eta_{j^\star}^T \mu| \leq \delta_0.$$

The following result characterizes a test for H_0 .

PROPOSITION 9.1. *The test which rejects when $F_{\delta_0, \sigma^2 \|\eta_{j^\star}\|^2}^{[\mathcal{V}^-, \mathcal{V}^+]}(\eta^T y) > 1 - \alpha$ is exact level α .*

PROOF. Let $\delta := \eta_{j^\star}^T \mu$. Define $T_{\delta_0} := \inf_{|\delta| \leq \delta_0} F_{\delta, \sigma^2 \|\eta_{j^\star}\|^2}^{[\mathcal{V}^-, \mathcal{V}^+]}(\eta^T y)$. Then:

$$\begin{aligned} \text{Type I error} &:= \sup_{|\delta| \leq \delta_0} \mathbb{P}_\delta(T_{\delta_0} > 1 - \alpha) \\ &\leq \sup_{|\delta| \leq \delta_0} \mathbb{P}_\delta \left(F_{\delta, \sigma^2 \|\eta_{j^\star}\|^2}^{[\mathcal{V}^-, \mathcal{V}^+]}(\eta^T y) > 1 - \alpha \right) \\ &= \alpha \end{aligned}$$

Next, we have that $T_{\delta_0} = F_{\delta_0, \sigma^2 \|\eta_{j^\star}\|^2}^{[\mathcal{V}^-, \mathcal{V}^+]}(\eta^T y)$, i.e., the infimum is achieved at $\delta = \delta_0$, so calculating T_{δ_0} is a simple matter of evaluating F_{δ_0} . This follows from the fact that F_δ is monotone decreasing in δ (c.f. Appendix A).

Finally, the Type I error is exactly α because the reverse inequality also holds:

$$\text{Type I error} \geq \mathbb{P}_{\delta_0}(T_{\delta_0} > 1 - \alpha) = \alpha.$$

□

Although the test is exact level α , the significance level of a test for a composite null is a “worst-case” Type I error; for most values of μ such that $|\eta^T \mu| \leq \delta_0$, the Type I error will be less than α , so the test will be conservative. Of course, what we lose in power, we gain in robustness to the assumption that $\eta^T \mu = 0$ exactly.

9.6. *How long a lasso should you use?* Procedures for fitting the lasso, such as **glmnet** (Friedman et al., 2010), solve (4.1) for a decreasing sequence of λ values starting from $\lambda_1 = \|X^T y\|_\infty$. The framework developed so far provides a means to decide when to stop along the regularization path, i.e.,

when the lasso has done enough “fitting.” In this section, we describe a path-wise testing procedure for the lasso,

The path-wise procedure is simple. At each value of λ :

1. Solve the lasso and obtain an equicorrelation set $\hat{E}_\lambda$ and signs $\hat{z}_{\hat{E}_\lambda}$.
2. Test $H_{0,\lambda} : X_{\hat{E}_\lambda}^T (I - P_{\hat{E}_\lambda})(\mu) = 0$ at level α . Rather than being conditional on only $(\hat{E}_\lambda, \hat{z}_{\hat{E}_\lambda})$, this test is conditional on the entire sequence of equicorrelation sets and signs $\{(\hat{E}^m, \hat{z}^m) = (E^m, z^m)\}$, as we describe below.

As λ decreases, we expect to reject the null hypotheses as the fit improves and stop once the first null hypothesis has been accepted.

To understand the properties of this procedure, we formalize it as a multiple testing problem. For each value $\lambda_1, \dots, \lambda_m$, we test H_{0,λ_i} . We test these hypotheses sequentially and stop after the first hypothesis has been accepted. Implicitly, this means that we accept all the remaining hypotheses.

Our next result shows that this procedure controls the family-wise error rate (FWER) at level α . Let V denote that number of false rejections. Then FWER is defined as $\mathbb{P}(V \geq 1)$. The practical implication of this result is the model selected by this procedure will be larger than the true model with probability α .

PROPOSITION 9.2. *The path-wise testing procedure controls FWER at level α .*

PROOF. Let $\hat{E}^m$ and $\hat{z}^m$ denote the complete sequence of equicorrelation sets and signs at $\lambda_1, \dots, \lambda_m$, i.e.,

$$\begin{aligned}\hat{E}^m &= \{\hat{E}_{\lambda_1}, \dots, \hat{E}_{\lambda_m}\} \\ \hat{z}^m &= \{\hat{z}_{\hat{E}_{\lambda_1}}, \dots, \hat{z}_{\hat{E}_{\lambda_m}}\}.\end{aligned}$$

We seek to control the family-wise error rate (FWER) when testing the hypotheses $H_{0,\lambda_1}, \dots, H_{0,\lambda_m}$, i.e., $\mathbb{P}(V \geq 1)$. We partition the space over all possible sequences $\hat{E}^m$ and $\hat{z}^m$:

$$\mathbb{P}(V \geq 1) = \sum_{(E^m, z^m)} \mathbb{P}\left(V \geq 1 \mid (\hat{E}^m, \hat{z}^m) = (E^m, z^m)\right) \mathbb{P}\left((\hat{E}^m, \hat{z}^m) = (E^m, z^m)\right).$$

Since $\sum_{(E^m, z^m)} \mathbb{P}\left((\hat{E}^m, \hat{z}^m) = (E^m, z^m)\right) = 1$, we can ensure $\text{FWER} \leq \alpha$ by ensuring

$$\mathbb{P}\left(V \geq 1 \mid (\hat{E}^m, \hat{z}^m) = (E^m, z^m)\right) \leq \alpha \text{ for any } (E^m, z^m).$$

Let λ_k denote the first λ_i for which H_{0,λ_i} is true. Then the event $V \geq 1$ is equivalent to the event that we reject H_{0,λ_k} because the preceding hypotheses $H_{0,\lambda_1}, \dots, H_{0,\lambda_{k-1}}$ are all false so we cannot make a false discovery before the k^{th} hypothesis. Thus

$$\mathbb{P}\left(V \geq 1 \mid (\hat{E}^m, \hat{z}^m) = (E^m, z^m)\right) = \mathbb{P}\left(\text{reject } H_{0,\lambda_k} \mid (\hat{E}^m, \hat{z}^m) = (E^m, z^m)\right).$$

Therefore, we can control FWER at level α by ensuring

$$\mathbb{P}\left(\text{reject } H_{0,\lambda} \mid (\hat{E}^m, \hat{z}^m) = (E^m, z^m)\right) \leq \alpha$$

for each $\lambda \in \{\lambda_1, \dots, \lambda_k\}$. $\square$

To perform a test of $H_{0,\lambda}$ conditioned on $\{(\hat{E}^m, \hat{z}^m) = (E^m, z^m)\}$, we apply the framework of Section 5. Let

$$\{A(E_i, s_i)y < b(E_i, s_i)\}$$

be the affine constraints that characterize the event $\{(\hat{E}_{\lambda_i}, \hat{z}_{\lambda_i}) = (E_i, z_i)\}$ from Proposition 4.2. The event $\{(\hat{E}^m, \hat{z}^m) = (E^m, z^m)\}$ is equivalent to the intersection of all of these constraints:

$$\underbrace{\begin{bmatrix} A(E_1, z_1) \\ \vdots \\ A(E_m, z_m) \end{bmatrix}}_{A(E^m, z^m)} y < \underbrace{\begin{bmatrix} b(E_1, z_1) \\ \vdots \\ b(E_m, z_m) \end{bmatrix}}_{b(E^m, z^m)}.$$

Now Theorem 5.2 applies, and we can obtain the usual pivot as a test statistic.

10. Conclusion. We have described a method for making inference about $\eta^T \mu$ in the linear model based on the lasso estimator, where η is chosen adaptively after model selection. The confidence intervals and tests that we propose are conditional on $\{(\hat{E}, \hat{s}) = (E, s)\}$. In contrast to existing procedures on inference for the lasso, we provide a pivot whose conditional distribution can be characterized exactly (non-asymptotically). This pivot can be used to derive confidence intervals and hypothesis tests based on lasso estimates anywhere along the solution path, not necessarily just at the knots of the LARS path as in Lockhart et al. (2014). Finally, our test is computationally simple: the quantities required to form the test statistic are readily available from the solution of the lasso.

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APPENDIX A: MONOTONICITY OF F

LEMMA A.1. *Let $F_\mu(x) := F_{\mu, \sigma^2}^{[a,b]}(x)$ denote the cumulative distribution function of a truncated Gaussian random variable, as defined as in (5.8). Then $F_\mu(x)$ is monotone decreasing in μ .*

PROOF. First, the truncated Gaussian distribution with CDF $F_\mu := F_{\mu, \sigma^2}^{[a,b]}$ is a natural exponential family in μ , since it is just a Gaussian with a different base measure. Therefore, it has monotone likelihood ratio in μ . That is, for all $\mu_1 > \mu_0$ and $x_1 > x_0$:

$$\frac{f_{\mu_1}(x_1)}{f_{\mu_0}(x_1)} > \frac{f_{\mu_1}(x_0)}{f_{\mu_0}(x_0)}$$

where $f_{\mu_i} := dF_{\mu_i}$ denotes the density. (Instead of appealing to properties of exponential families, this property can also be directly verified.)

This implies

$$f_{\mu_1}(x_1)f_{\mu_0}(x_0) > f_{\mu_1}(x_0)f_{\mu_0}(x_1) \quad x_1 > x_0.$$

Therefore, the inequality is preserved if we integrate both sides with respect to x_0 on $(-\infty, x)$ for $x < x_1$. This yields:

$$\begin{aligned} \int_{-\infty}^x f_{\mu_1}(x_1)f_{\mu_0}(x_0) dx_0 &> \int_{-\infty}^x f_{\mu_1}(x_0)f_{\mu_0}(x_1) dx_0 & x < x_1 \\ f_{\mu_1}(x_1)F_{\mu_0}(x) &> f_{\mu_0}(x_1)F_{\mu_1}(x) & x < x_1 \end{aligned}$$

Now we integrate both sides with respect to x_1 on (x, ∞) to obtain:

$$(1 - F_{\mu_1}(x))F_{\mu_0}(x) > (1 - F_{\mu_0}(x))F_{\mu_1}(x)$$

which establishes $F_{\mu_0}(x) > F_{\mu_1}(x)$ for all $\mu_1 > \mu_0$. $\square$

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